

An algebra of analogies

An analogy problem of the form $A : B :: C : D$ asks a solver to extract a change from a demonstrated pair and to apply it at a target. Hofstadter and Mitchell (1995) studied the letter-string version through the frequencies of Copycat’s answers over many runs; Chollet (2019) poses the grid version in the Abstraction and Reasoning Corpus as a test of skill acquisition from few examples. I am interested in the algebra a domain must carry for extraction and application to be well posed, and the algebraic form each variation from it takes in the solver’s returned answers.

An analogical domain is a quadruple $\langle \mathcal{O}, \text{Part}, \mathcal{B}, \kappa \rangle$ of observations, a parser returning finitely many parses of each observation, a basis of binary relations on parsed observations, and a cost on the theories built from the basis. A theory T explains a pair when the pair lies in its semantics under some parses, and it predicts at a target C the set $\text{Ans}_T(C)$ of observations it relates to C . The survivors are the theories explaining the demonstrated pair and predicting at least one answer at the target. Resolution returns the union of the predictions of the cheapest survivors. For example, on an alphabet ending at z , successor on the final letter is the cheapest explanation of $abc \rightarrow abd$, and has an empty answer set at xyz .

The explanations form a transformation category Γ , whose arrows from A to B are the behaviors of theories explaining (A, B) . In the functional fragment, a theory is a word in the generators of a monoid acting on the parsed observations; two words with one value are one arrow. Under the commonality reading, the proportion $A : B :: C : D$ holds when one cheapest theory explains both pairs; Antić (2022) develops this reading in universal algebra. Under the difference reading, the proportion holds when $\Gamma(A, B)$ and $\Gamma(C, D)$ share an arrow. Over a group action the difference reading is the commonality reading, restricted to graphs of single group elements; admitting unions widens the commonality reading strictly. Every two proportional pairs share exactly one graph theory if and only if the action is free.

Structure-mapping ranks the alignments of two situations by systematicity (Gentner, 1983), and Murena et al. (2020) resolve letter-string proportions by the transformation of least description length; both rank a set of rival explanations. The algebra describes this set: whether it has one element, how many it has when the source is symmetric, when it is empty, and when every symmetric ranking returns at least two of its elements.

Classification

Well-posedness of the difference reading asks that every pair have an explanation, that extraction return one arrow, and that every arrow act at every object; I call these conditions existence, uniqueness, and applicability. Existence and uniqueness entail invertibility: a round trip $A \rightarrow B \rightarrow A$ lies in the vertex monoid at A , which uniqueness makes the singleton $\{e\}$. All three conditions hold exactly when the parsed observations form a torsor: the arrows form a group acting on them freely and transitively. On a torsor, the difference reading is the equation $\delta(A, B) = \delta(C, D)$, and the heap operation $(A, B, C) \mapsto \delta(A, B) \cdot C$ completes every proportion uniquely. The three conditions are independent, each failing with the other two intact: applicability on an alphabet ending at z ; existence for two asymmetric figures under the quarter turns, a free action with two orbits; uniqueness for the four vertices of a square under the dihedral group D_4 , where every hom-set holds two arrows.

Over an abelian group, completion is the classic parallelogram completion, $D = C + (B - A)$, which is the model of Rumelhart and Abrahamson (1973) and the form of the linear analogies in word embeddings (Allen and Hospedales, 2019). Baker (2023) takes the torsor equation as the definition of proportion, and shows that proportion spaces, torsors, and heaps are equivalent. In the classification, the torsor is the conclusion of a theorem, from which we observe, by what condition or conditions it fails, the other domains.

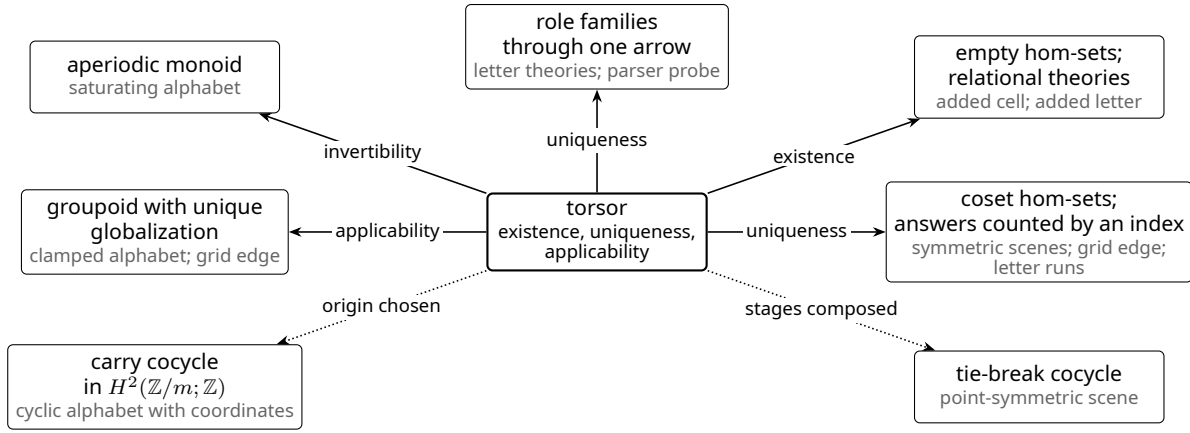


Figure 1: Variations on the torsor. A solid arrow leads to the structure that results when its labeled property fails; a dotted arrow leads to structure added while every condition holds. Note that “grid edge” appears twice, because the edge probe combines a stabilizer coset with a partial action.

Variations

Restricting the integer translations to the window of letters a through z yields a partial action: $+1$ is undefined at z, while every pair of letters maintains its difference. The arrows form a groupoid with singleton hom-sets, and the partial action has a unique globalization: the integers translating the line (Kellendonk and Lawson, 2004). Up to equivariant bijection, the clamped alphabet determines the line. Knowledge of clocks xya replaces it by the cyclic alphabet, which is a change of domain equipped with modular arithmetic.

The quotient by $m\mathbb{Z}$ yields a cyclic alphabet of m letters, a $\mathbb{Z}/m$ -torsor. Assigning integer coordinates to its letters amounts to choosing a section λ of $q: \mathbb{Z} \rightarrow \mathbb{Z}/m$. The carry of longhand addition, for example, (Isaksen, 2002),

$$c(\bar{x}, \bar{y}) = (\lambda(\bar{x}) + \lambda(\bar{y}) - \lambda(\overline{x+y})) / m,$$

is a 2-cocycle, whose class generates $H^2(\mathbb{Z}/m; \mathbb{Z}) \cong \mathbb{Z}/m$ (Brown, 1982). Because the class is nonzero, every choice of origin carries the defect. Tested on coordinates, the proportion $y : b :: c : f$ fails by one full turn, since both circle differences are $+3$, while the first integer difference is $1 - 24 = -23$.

Saturation completes a translation that would leave the window by the nearest-point retraction onto it, and the resulting maps form an aperiodic monoid of clamped translations (Schützenberger, 1965). Since existence holds, the loss of invertibility forces a failure of uniqueness: $\Gamma(a, c)$ holds the translation $+2$, the constant map at c , and the clamps between them.

When a group acts and $t \cdot A = B$, the explanations of (A, B) form the coset $t \text{Stab}(A)$. The answers at a target C form the image under t of the $\text{Stab}(A)$ -orbit of C :

$$\text{Ans}(C) = t \text{Stab}(A) \cdot C, \quad |\text{Ans}(C)| = [\text{Stab}(A) : \text{Stab}(A) \cap \text{Stab}(C)].$$

The answer is unique when every symmetry of the source fixes the target. Both quarter turns explain the pair formed by a point-symmetric scene and its quarter turn, and they return two answers at an asymmetric target, and one at a target sharing the half-turn symmetry. Partial generators prune a coset, which makes the index an upper bound on the number of answers.

Strings of length n carry the same structure under the group $\Sigma_n \cong \mathbb{Z}/26 \rtimes (\mathbb{Z}/2 \times \mathbb{Z}/2)$, generated by translation, reversal, and negation of the alphabet. A run of step d with $2d \not\equiv 0 \pmod{26}$ has a stabilizer of order two, whose nontrivial element is the half turn about the run’s midpoint in the plane of positions and letters. By the coset criterion, the correspondences carrying abc onto xyz form a coset of this stabilizer, and they carry abd to xya and to wyz . At kji the corresponding coset returns kjh and lji . On the clamped alphabet the translation carries abd out of the window, and wyz remains. Copycat’s answer wyz , thus, completes the central permutation $A : C :: B : D$ through a symmetry of the source run.

Existence fails between a scene and its copy with one added cell, since rigid motions preserve the number of cells. An indeterminate copy explains this pair as a relation; determinate copies, which adjoin the images of

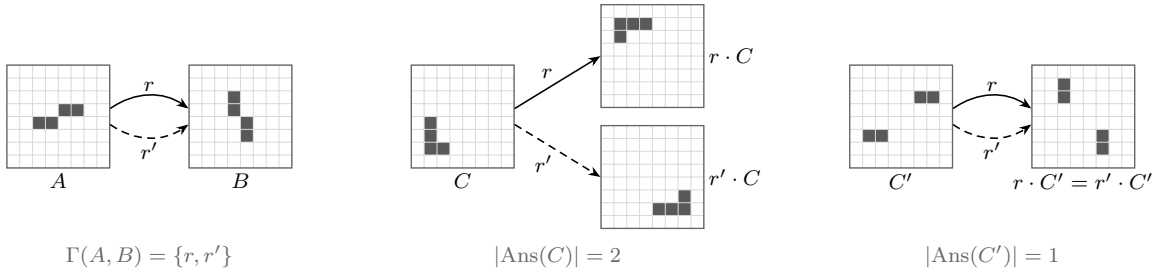


Figure 2: Answers from a coset of explanations. The group D_4 acts on the grid about its center, with r the clockwise quarter turn and $r' = r^3$ the counterclockwise one. The source A has stabilizer $\{e, r^2\}$, and $B = r \cdot A$. Both quarter turns carry A to B and return two answers at the asymmetric target C . At C' , which the half turn fixes, their answers coincide. Solid arrows apply r , and dashed arrows apply r' .

the marked cells under fixed motions, explain it through a coset of the stabilizer of the mark.

Selection

Reweightings reorders the spellings of one arrow and leaves the answer fixed, as with three successors and twenty-three predecessors on the cyclic alphabet. Distinct arrows in one coset are semantic rivals, whose values agree at the source and can differ at the target.

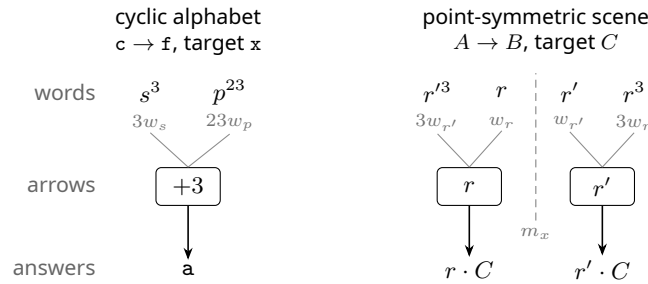


Figure 3: On the cyclic alphabet, every word explaining $c \rightarrow f$ evaluates to the arrow $+3$. Reweighting the successor s against the predecessor p reorders these words and fixes the answer at x . At the point-symmetric scene, the words explaining $A \rightarrow B$ evaluate to r or r' , which return different answers at C . Conjugation by the reflection m_x , drawn as the dashed axis, exchanges the words over r with those over r' and w_r with $w_{r'}$. A cost invariant under reflections, therefore, ties r with r' .

Role families are theories outside the group: functions on parsed observations agreeing at the source and parting at the target. On letters, successor on the final letter, successor at the third position, successor on the final block, and the advance of a run's final term by its step all carry abc to abd , and they return three different strings at $iiijkk$.

Several demonstrations (A_i, B_i) constrain a single theory; in a group action, their explanations form the intersection of the cosets $t_i \text{Stab}(A_i)$, which is empty or a coset of $K = \bigcap_i \text{Stab}(A_i)$. The answers at C then number $[K : K \cap \text{Stab}(C)]$. The count is nonincreasing as demonstrations accumulate; it equals one exactly when every symmetry common to the sources fixes the target.

Let a group Q preserve parser, basis, and cost, and let it fix the problem. An equivariant choice of one minimizer exists exactly when Q fixes some minimizer. Wherever two blocks cost less than three single letters, reversal fixes aba and exchanges its two cheapest segmentations, $[ab][a]$ and $[a][ba]$. Hence, every deterministic segmenter breaks reversal symmetry on aba . Frame averaging in equivariant networks likewise replaces a canonical group element by an equivariant set of group elements (Puny et al., 2022).

Over a group action, the stage hom-sets compose to the direct one, $\Gamma(B, C) \circ \Gamma(A, B) = \Gamma(A, C)$, and every chained selection lies in the direct coset. Selection still fails to compose: on a point-symmetric source with

stabilizer $S = \{e, r^2\}$, a deterministic tie-break is a section λ of $D_4 \rightarrow D_4/S$, and the chained and direct selections along two stages differ by $c(\bar{g}_2, \bar{g}_1) = \lambda(\bar{g}_2) \lambda(\bar{g}_1) \lambda(\bar{g}_2 \bar{g}_1)^{-1} \in S$. This c is a 2-cocycle, whose class is the class of the central extension $1 \rightarrow S \rightarrow D_4 \rightarrow D_4/S \rightarrow 1$. The class is nonzero because every subgroup of D_4 of order four contains r^2 , the square of a quarter turn. Every deterministic tie-break, therefore, diverges from direct resolution on the chain $A \rightarrow r \cdot A \rightarrow A$. On letters, the corresponding extension splits, since the correspondences without reversal form a complement to every run stabilizer. Selecting those correspondences composes along every chain. The carry and the tie-break defect are classes of one kind, each obstructing a homomorphic choice of representatives for a quotient.

Consequences

I designed the grid probes to vary one input of resolution each, from the symmetry of the source to expressibility. Recomputed in the group $\mathbb{Z}^2 \rtimes D_4$ of lattice motions, the weight probe and the edge probe turned out to be stabilizer probes as well. Their sources are symmetric about centers away from the center of the grid, and their hom-sets are cosets with eight and four elements. With determinate copies, the expressibility probe becomes a coset probe too, and the parser probe keeps a trivial stabilizer. Each probe isolates one input; over a group action, reweighting and exclusion at a boundary operate within a coset.

For letters, four theories came from the demonstration, the reversal pairing from *xyz*, and the run generator from the answers *kjh*, *mrrjjjj*, and *aababcabcde*. I classified the targets *kji* and *iijjkk* after fixing these theories. At the default weights, the scheme returns *kjj*, *iijjkl*, and *mrrjjk*, where Copycat most often returns *kjh*, *iijjll*, and *mrrkkk* (Hofstadter and Mitchell, 1995). The scheme returns Copycat’s modal answers at all four targets in one region of weight space once a stipulation on the run generator is dropped, and three of the four with the stipulation kept. The hypothesis that answer distributions are images of distributions over weights holds with this region as its domain. The scheme also returns strings outside the corpus, among them *xza* and *hhjjkk*, and the corpus contains *yyz*, *dyz*, and *dji*, which lie outside the difference reading and inside the commonality reading through unions of conjugate theories whose cost is unassigned.

If a population of human solvers is a population of weightings and tie-breaks over one domain, answers to the point-symmetric probe should split between the two quarter turns. Presenting each problem beside its mirror image distinguishes a fixed preference for clockwise turns from an equivariant tie-break. Under the fixed preference both answers turn the same way; under the equivariant tie-break they are mirror images. Presented as a chain of stages, a problem should draw its answers from the direct answer set, and a fixed tie-break applied at three quarter-turn stages selects $r^3 = r'$ where the direct problem selects r . Output accuracy misleads wherever two role families agree at the test input, since a correct output then fits both. Beger et al. (2025) find that, in the textual modality, accuracy on ConceptARC tasks overstates how often the rule a model states captures the intended abstraction.

Relations

Beyond the functional fragment a theory is a relation, as the union theories of the commonality reading and the covering relation of *hand : glove :: foot : shoe* are. Stating such theories with a cost requires a calculus of relations. Bonchi et al. (2024) give a complete axiomatization of a diagrammatic calculus of relations typed by arity, with the expressive power of first-order logic. With Avi Craimer and Nathan Haydon, I am developing a single-sorted counterpart over a domain carrying an injective pairing; its operations are composition, converse, complement, and a monoidal product, and a copying constant satisfies the Frobenius laws. The standard models are proper quasi-projective relation algebras in the sense of Tarski and Givant (1987). Through their translation of first-order logic, the calculus is semantically as expressive as first-order logic with equality.

Soundness is proved, and completeness for the standard semantics is open. By the argument of Monk (1964), the equations valid in the relation-algebra reducts of the standard models are nonfinitely axiomatizable. Tarski and Givant’s quasi-projective relation algebras form a finitely axiomatized class, and every member of it is representable. Adding quasi-projections to the language lifts the obstruction. The standard models also satisfy the unicity of codes, by which a code is determined by its two components, and a quasi-projective relation

algebra can violate it. A complete axiomatization must yield this condition, along with Tarski and Givant’s axioms.

The plan is to write the theories of resolution as terms of this calculus, with cost read off term size, and to select abstractions in the relational fragment. There, an abstraction Θ is a finite graph of roles whose edges carry relations from a finite library Λ , and an interpretation of a situation X under a partition P is a relation-preserving map $\iota: \varepsilon_\Lambda(X, P)/\sim \rightarrow \Theta$ from the parts, enriched with their Λ -relations and quotiented by structural indistinguishability. Selection minimizes description length (Rissanen, 1978),

$$(\Theta^*, \{P_i, \iota_i\}^*) = \arg \min_{\Theta, \{P_i, \iota_i\}} \left[L_\Lambda(\Theta) + \sum_i L_\Lambda(P_i, \iota_i) \right],$$

subject to every demonstration and the target lifting into Θ . With the interpretation term set to zero and L_Λ taken as κ , the criterion is resolution. The interpretation term would price the preference among parses, which the letter computations apply as an unpriced convention.

Agenda

The commonality reading needs a classification of its own, with a shared theory for every two pairs, a unique shared graph, and applicability of the shared theory at a new target in the places of existence, uniqueness, and applicability. For the relational fragment, the question is whether the explanations of a pair (written as terms of the calculus) carry structure comparable to the stabilizer coset. The cost model is open as well, since charging the word executed at the target in place of the theory explaining the source ranks the same theories differently. On the logical side, I will investigate whether the present axioms derive Tarski and Givant’s axioms and the unicity of codes, and whether a representation argument then yields completeness.

On the empirical side, an initial study fits weights and tie-breaks to human answer distributions on letter strings and grid scenes, concentrating on problems where the algebra returns a coset or a tie, and tests the predictions about tie-breaks with mirrored and chained presentations. Mental rotation latencies grow with the angle of rotation (Shepard and Metzler, 1971), which gives an independent estimate of the cost of a turn. A benchmark built from the rival sets would pair test inputs on which two role families agree with inputs on which they part, and output accuracy and the selected theory would then come apart by construction. Liu et al. (2023) relate the depth at which transformers simulate an automaton to its Krohn–Rhodes decomposition (Krohn and Rhodes, 1965), in which the cyclic alphabet’s transition monoid is a group and the saturating alphabet’s is aperiodic. I plan to train sequence models on these two alphabets, matched in size and generators, and to compare how proportion completion generalizes on each. Learning the library Λ across a corpus by total description length would bring the library learning of DreamCoder (Ellis et al., 2021) to role-typed relations.

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